

PROFESSIONAL EXPERIENCE

Addgene - Data Engineer II - Backend Platform

Mar 2025 - Present

Python | Django | AWS (S3, Lambda, Step Functions, DynamoDB) | BigQuery | Looker | Spark | Celery | dbt | Elasticsearch

- Owned the architecture, implementation, and production rollout of a **shipment visibility platform** for **200K** annual shipments, delivering **reshipment analytics** identifying ~\$**1.54M** in reshipment costs across **14K** orders.
- Built **distributed, fault-tolerant ETL pipelines** ingesting terabytes of scientific data from **NCBI, S3, and PostgreSQL** into **BigQuery**, implementing retry mechanisms and automated validation to power inventory and order analytics.
- Engineered a **SEGUID-based DNA sequence hashing algorithm** for automated validation, reducing processing latency from **8 hours to ~2 minutes (99.6% improvement)**.
- Architected an Elasticsearch indexing pipeline enabling **low-latency search** and filtering across **300K+** catalog records.
- Improved **application and query performance by 20%** through ORM profiling, query refactoring, BigQuery clustering and materialized views.

Autharva Inc - Software Integrations Engineer

Feb 2025 - Mar 2025

Python | Grafana | AWS CloudWatch | Monitoring & Observability | Root Cause Analysis

- Resolved Workday-to-S3 data ingestion failures, **reducing pipeline failures by 90%** through root-cause analysis and workflow optimization.
- Built **Grafana** dashboards and **CloudWatch** alerts to improve **production reliability**, monitoring over **50,000 entitlements**, enabling security to **flag 40+ over-privileged accounts**.

Blusmart Mobility - Backend Engineer

Feb 2022 - Jan 2023

Java | REST/gRPC | Selenium | Spring Boot | Microservices | Redis | Docker | ELK Stack

- Designed, implemented, and deployed scalable **Spring Boot microservices** and **REST/gRPC services** supporting **1.47M+ users** and **20K+ daily rides**, delivering reliable backend services for ride booking and fleet operations.
- Built **production observability** using the **ELK stack** to monitor distributed services, analyze logs, and accelerate incident detection and root-cause analysis, reducing production issues.
- Optimized backend performance using **Redis** caching for frequently accessed data.

Coforgetech - Software Engineer

Oct 2020 - Feb 2022

Java | Spring Boot | SQLite | Postman | JWT Authentication | OOP | TDD

- Built core meal-planning services serving **2,000+ catering staff**, maintaining **sub-200 ms p95 latency** and stable performance under peak usage.
- Implemented **JWT-based authentication** and role-based access control, **securing API endpoints** and simplifying authorization across user roles.
- Performed end-to-end **API testing**, optimized backend queries to **improve API response times** and scalability under concurrent load.

Genpact Headstrong -Software Engineer

Dec 2019 - Sept 2020

- Built **Python-based test automation** for **unit, functional, and regression testing** reducing downstream QA defects by ~**50%**.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, Go, Bash

Backend & Distributed Systems: Spring Boot, Django, Node.js, REST APIs, gRPC, GraphQL, OpenAPI, Microservices, Celery, TDD

Cloud & Infrastructure: AWS, [GCP](#), Snowflake, [Azure](#), Docker, Kubernetes, Jenkins, Airflow, AWS Step Functions

Data & Storage: Kafka, Elasticsearch, DynamoDB, Redis, PostgreSQL, MySQL, BigQuery, Spark, dbt

Observability & Reliability: Grafana, Prometheus, CloudWatch, ELK, Splunk, monitoring, alerting, root-cause analysis

MLOps/AI: RAG pipelines, Vector databases, Embeddings, [LLM Driven Development](#)

PROJECTS

[Logitt Application](#) | ReactJS, Java, Spring Boot, JPA, AWS, Hibernate ORM, TDD

- Built and deployed a **full-stack personal finance application** using ReactJS, Java, Spring Boot, MySQL and AWS.
- Created a dashboard and analytics features that helped users identify spending patterns and reduce unnecessary expenses by up to 40%.

[AskMyDocs RAG Pipeline](#) | Python, LangChain, ChromaDB, Transformers, SentenceTransformers, Groq, Vector Embeddings, NLP

- Built an end-to-end RAG pipeline over an NLP corpus using **LangChain, ChromaDB, Transformers, BERT, GPT-3, SentenceTransformers (all-MiniLM-L6-v2)**, and **Groq LLMs**.
- Implemented **PDF chunking, embedding generation, cosine similarity retrieval** and **citation-backed answering**.

[Medical Specialty Classifier Using NLP](#) | Python, Streamlit, Random Forests, SVM, XGBoost, Vector Embeddings, Pandas, Numpy, scikit-learn

- Trained and fine-tuned ML models and other deep learning models like **LSTMs** and **RoBERTa**, achieving 69% accuracy.
- Conducted data **cleaning, tokenization, stemming**, and **stop-word removal** and generated **TF-IDF, Word2Vec, ClinicalBERT** embeddings.

EDUCATION

Northeastern University, Boston, MA, USA - Masters in Computer Science

Jan 2023 - Dec 2024